required hypothesis here and in subsequent uses, including Lemma 2.25 and Corollary 2.30. The contrary permission at the opening of Section 6 is removed.

Lemma 2.24. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc. Then $\mathfrak{a}$ is t-atomic for all $t \in \mathrm{RV}$.*

Proof. For the case $n = 1$, if $\mathfrak{a}$ were not t-atomic then, by Corollary 2.21 and Lemma 2.13, it would contain a point algebraic over the fixed ground base. This point belongs to a finite ground-definable set. Atomicity would then force the whole infinite open disc $\mathfrak{a}$ to lie in that finite set, a contradiction. For the case $n > 1$, by induction on n, $\mathfrak{a}_1$ is t-atomic and for every $a \in \mathfrak{a}_1$, $\mathfrak{a}_{>1}$ is (a, t)-atomic. So $\mathfrak{a}$ cannot have a nonempty proper t-definable subset. □

Partly resolved referee R029 [Gap] – Atomicity proof: algebraic versus definable.
Original source line 886. Lemma 2.13 only gives an algebraic point over the original base, not necessarily a definable point. This suffices: an infinite atomic disc cannot meet a finite base-definable set. Replace 'definable point' at this step by that argument. The final sentence also needs 'nonempty proper t-definable subset'. Finally, indicate why Corollary 2.21 remains applicable whenever previously named disc parameters occur in the base; its stated hypothesis excludes such extra parameters.

Author resolution. The author approves both proof corrections: the point is algebraic over the fixed ground base, so atomicity would place the infinite disc inside a finite ground-definable set; and the final sentence now says "nonempty proper." The statement is unchanged. The separate question about applicability over additional disc-imaginary ground parameters remains open.

Lemma 2.25. *Suppose that $\Gamma(\mathbb{S})$ is nontrivial. Let $A \subseteq \mathrm{VF}^n$ be a set that generates a type and $\mathfrak{b} \subseteq A$ an open (or closed) polydisc. Then $\mathfrak{b}$ is $\ulcorner \mathfrak{b} \urcorner$-atomic.*

Proof. We do induction on n. The base case $n = 1$ is just [18, Lemma 3.4]. For the case $n > 1$, write $\mathfrak{b} = \prod_i \mathfrak{b}_i$, $\mathfrak{b}' = \prod_{i>1} \mathfrak{b}_i$, and $\gamma = \mathrm{rad}(\mathfrak{b}_1)$. For any $a \in \mathfrak{b}_1$, A_a generates a type. So, by Lemma 2.23 and the inductive hypothesis, $\mathfrak{b}'$ is $(a, \gamma, \ulcorner \mathfrak{b}' \urcorner)$-atomic and hence $\ulcorner \mathfrak{b} \urcorner$-atomic. Similarly $\mathfrak{b}_1$ is $\ulcorner \mathfrak{b} \urcorner$-atomic. It follows that $\mathfrak{b}$ must be $\ulcorner \mathfrak{b} \urcorner$-atomic. □

Lemma 2.26. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc and $f : \mathfrak{a} \longrightarrow \mathcal{P}(\mathrm{VF})$ a definable finitary function such that $\mathrm{rv} \restriction f(a)$ is injective for every $a \in \mathfrak{a}$. Then $\mathrm{rv} \circ f$ is constant.*

Proof. This is clear since, by Lemma 2.24, for every $t \in \mathrm{rv}(\bigcup f(\mathfrak{a}))$, $\mathfrak{a}$ is t-atomic and hence f induces a t-definable function $\mathfrak{a} \longrightarrow t^\sharp$. The lemma follows. □

A *thin annulus* is a set of the form $\mathfrak{c}(a, \gamma) \setminus \mathfrak{o}(a, \gamma) = \{x \in \mathrm{VF} : \mathrm{val}(x - a) = \gamma\}$, where $a \in \mathrm{VF}$ and $\gamma \in \Gamma$. Equivalently, it is a closed disc with one maximal open subdisc removed.

Lemma 2.27 ([18, Lemma 3.8])**.** *Let $A \subseteq \mathrm{VF}$ be an atomic open disc or an atomic closed disc or an atomic thin annulus. Let $f : A \longrightarrow \mathrm{VF}$ be a definable function. Then $f(A)$ is also atomic of one of these three forms.*

Resolved referee R163 [Clarification] – Define thin annuli before using them.
Original source line 907. The term "thin annulus" has not been defined in this manuscript. Specify a closed disc with one maximal open subdisc removed; after choosing the center of the removed subdisc, it is a shell $\{x : \mathrm{val}(x - a) = \gamma\}$. This makes the alternatives here and the later component argument unambiguous.

Author resolution. The author approves the finite-radius shell definition $\mathfrak{c}(a, \gamma) \setminus \mathfrak{o}(a, \gamma) = \{x \in \mathrm{VF} : \mathrm{val}(x - a) = \gamma\}$, with $\gamma \in \Gamma$. It is now stated before the first use in Lemma 2.27.

Lemma 2.28 ([18, Lemma 3.13])**.** *Let $A \subseteq \mathrm{VF}$ be an atomic open disc and $f : A \longrightarrow \mathrm{VF}$ a definable function. Then either f is constant or $f(A)$ is also an atomic open disc.*

Lemma 2.29. *Let $\mathfrak{a}$, f be as in Lemma 2.26 and suppose that f is not constant. Then there are $t_1, \ldots, t_k \in \mathrm{RV}_\infty$ and for each i, a t_i-definable function $f_i : \mathfrak{a} \longrightarrow t_i^\sharp$ such that $f = \bigcup_i f_i$ and each $f_i(\mathfrak{a})$ is either a point or a t_i-atomic open disc.*

Proof. By Lemma 2.26 and injectivity of rv on each finite fiber, f decomposes into finitely many t-definable single-valued branches $g : \mathfrak{a} \longrightarrow t^\sharp$. A branch with $t = \infty$ is the constant zero branch. Any constant branch already has the required form, so fix a nonconstant branch, for which $t \in \mathrm{RV}$. Lemma 2.24 makes $\mathfrak{a}$ atomic over t.

Every coordinate line of $\mathfrak{a}$ is atomic over its fixed coordinates and t. By Lemma 2.28, its image under g is a point or an open disc. Any two points of $\mathfrak{a}$ can be joined by changing one coordinate at a time. Starting at a point with image p and ending at a point with different image, the first nonconstant line image is an open neighborhood of p contained in $g(\mathfrak{a})$. Thus $g(\mathfrak{a})$ is open. It is a proper definable subset of VF, since it lies in $t^\sharp$.

By C-minimality, every point of a proper definable open subset of VF lies in a maximal open disc contained in that subset. Choose such a disc $D \subseteq g(\mathfrak{a})$ containing one value of g. An open line image that meets D is contained in D: intersecting open discs are nested, and maximality excludes a larger one inside $g(\mathfrak{a})$. A point image also stays in D when its initial value is in D. The coordinate chains therefore keep every value of g in D, so $g(\mathfrak{a}) = D$. The definable image of an atomic set is atomic over the same base, hence this open disc is t-atomic, as required. □

Resolved referee R030 [Gap; revised] – A unary image lemma is used on a polydisc.
Original source line 918. For $n > 1$, Lemma 2.27 does not directly apply to $f : \mathfrak{a} \longrightarrow \mathrm{VF}$, since it has a unary domain. Supply the missing argument that $f(\mathfrak{a})$ has just one primitive component. A possible route is the canonical finite component decomposition of an atomic unary set: every coordinate slice of $\mathfrak{a}$ is atomic over its fixed coordinates, and its image is a point or an open disc by induction. Show that such an image must lie in one primitive component, then connect any two source points by changing coordinates one at a time. The same coordinate-chain argument excludes a closed disc or thin annulus as the whole image, because each slice image lies in a single maximal open subdisc. This fills the role of the inapplicable unary citation, once the component claim is spelled out.

Author resolution. The author approves the coordinate-chain proof. Unary coordinate-line images are points or open discs; maximal open discs in a proper definable open image then force the entire nonconstant image to be one open disc. This replaces the inapplicable unary citation on the polydisc. The tag ranges here and in Corollary 2.30 are also enlarged to RV_∞, retaining the constant zero branch.

Corollary 2.30. *Let $A \subseteq \mathrm{VF}^n$ be a definable set and $f : A \longrightarrow \mathcal{P}(\mathrm{VF})$ a definable finitary function as in Lemma 2.26. Then there are a definable finite partition $(A_i)_i$ of A and, for each i and all open polydiscs $\mathfrak{a} \subseteq A_i$, finitely many $t_{\mathfrak{a}j}$-definable functions $f_{\mathfrak{a}j} : \mathfrak{a} \longrightarrow t_{\mathfrak{a}j}^\sharp$, where $t_{\mathfrak{a}j} \in \mathrm{RV}_\infty$, such that $f \restriction \mathfrak{a} = \bigcup_j f_{\mathfrak{a}j}$ and each $f_{\mathfrak{a}j}(\mathfrak{a})$ is either a point or an open disc.*

Proof. For $a \in A$, let $D_a \subseteq A$ be the type-definable subset containing a. By Lemma 2.25, every open polydisc $\mathfrak{a} \subseteq D_a$ is $\ulcorner \mathfrak{a} \urcorner$-atomic and hence, by Lemma 2.28, the assertion holds for $\mathfrak{a}$. Then, by compactness, the assertion must hold in a definable subset $A_a \subseteq A$ that contains a; by compactness again, it holds in finitely many definable subsets $A_1, \ldots, A_m$ of A with $\bigcup_i A_i = A$. Then the partition of A generated by $A_1, \ldots, A_m$ is as desired. □

Referee R031 [Error] – The local branch functions need the disc parameter.
Original source line 925. The functions $f_{\mathfrak{a}j}$ must be allowed parameters $(\ulcorner\mathfrak{a}\urcorner, t_{\mathfrak{a}j})$, not just $t_{\mathfrak{a}j}$. Their domains are $\mathfrak{a}$, so literal $t_{\mathfrak{a}j}$-definability would imply that every such disc is $t_{\mathfrak{a}j}$-definable. Already for $f(a) = \{1\}$ on $A = \mathrm{VF}$, this is impossible for arbitrary subdiscs, since $t_{\mathfrak{a}j} = \mathrm{rv}(1)$ is fixed. The proof itself introduces $\ulcorner\mathfrak{a}\urcorner$, which explains the missing parameter. Also replace the appeal to the unary Lemma 2.28 by the corrected multidimensional Lemma 2.29, and explicitly formulate the uniform first-order property used in the compactness step.

Definition 2.31. Let $f : A \longrightarrow B$ be a bijection between two sets A and B, each with exactly one VF-coordinate. We say that f is *concentric* if, for every VF-fiber f_t of f and all open polydiscs $\mathfrak{a} \subseteq \mathrm{dom}(f_t)$, $f_t(\mathfrak{a})$ is also an open polydisc; if both f and f^{-1} are concentric then f has the *disc-to-disc property*.

Lemma 2.32 ([18, Proposition 3.19])**.** *Let $f : A \longrightarrow B$ be a definable bijection between two sets A and B, each with exactly one* VF*-coordinate. Then there is a definable finite partition $(A_i)_i$ of A such that every restriction $f \restriction A_i$ has the disc-to-disc property.*

2.3. Continuity and fiberwise properties.

Definition 2.33. We set $\dim_{\mathrm{RV}}(\varnothing) = -\infty$. For a nonempty definable subset $U \subseteq \mathrm{RV}_\infty^m$, the RV*-dimension*, denoted by $\dim_{\mathrm{RV}}(U)$, is the smallest number k such that there is a definable finite-to-one function $f : U \longrightarrow \mathrm{RV}^k$ (RV^0 is taken to be the singleton $\{\infty\}$).

Resolved referee R164 [Clarification; new] – Dimension of the empty set.
Original source line 941. Specify $\dim_{\mathrm{VF}}(\varnothing) = \dim_{\mathrm{RV}}(\varnothing) = -\infty$ (or another negative convention) separately. The VF definition otherwise has no largest natural number for the empty set, whereas the literal RV definition gives it dimension 0. With that latter convention the boundary lemma fails when $n = 0$, since its boundary is empty but $0 < 0$ is false. Alternatively restrict that lemma to $n \geq 1$, while still specifying the empty VF dimension.

Author resolution. The author approves $\dim_{\mathrm{VF}}(\varnothing) = \dim_{\mathrm{RV}}(\varnothing) = -\infty$. Both definitions now state this convention separately and retain their original clauses for nonempty sets, including zero-coordinate cases.

Definition 2.34. Let A be a subset of VF^n. The RV*-boundary* of A, denoted by $\partial_{\mathrm{RV}}A$, is the definable subset of $\mathrm{rv}(A)$ such that $t \in \partial_{\mathrm{RV}}A$ if and only if $t^\sharp \cap A$ is a proper nonempty subset of $t^\sharp$. The definable set $\mathrm{rv}(A) \smallsetminus \partial_{\mathrm{RV}}A$, denoted by $\mathrm{int}_{\mathrm{RV}}(A)$, is called the RV*-interior* of A.

Referee R032 [Typo] – Definability of the RV-boundary.
Original source line 945. Insert 'definable' before 'subset of VF^n'; without that assumption the boundary and interior need not be definable.

Obviously, $A \subseteq \mathrm{VF}^n$ is an RV-pullback if and only if $\partial_{\mathrm{RV}}A$ is empty.

Lemma 2.35. *Let A be a definable subset of* VF^n*. Then* $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}}A) < n$.

Proof. We proceed by induction on n. The base case $n = 1$ follows immediately from C-minimality since $t^\sharp$ contains a t-definable proper subdisc for every $t \in \partial_{\mathrm{RV}}A$.

For the inductive step, since $\partial_{\mathrm{RV}}A_a$ is finite for every $i \in [n]$ and every $a \in \mathrm{pr}_{\tilde{i}}(A)$, by Corollary 2.21 and compactness, there are a definable finite partition $(A_{ij})_j$ of $\mathrm{pr}_{\tilde{i}}(A)$ and, for each A_{ij}, a definable finitary function $f_{ij} : A_{ij} \longrightarrow \mathcal{P}(\mathrm{VF})$ such that for all $a \in A_{ij}$, $\mathrm{rv}(f_{ij}(a)) = \partial_{\mathrm{RV}}A_a$ if A_a is not an RV-pullback or $f_{ij}(a) = \{0\}$ otherwise. By Corollary 2.30, we may assume that if

$t^\sharp \subseteq A_{ij}$ then the restriction $\mathrm{rv} \upharpoonright f_{ij}(t^\sharp)$ is constant. Hence each f_{ij} induces a definable finitary function $C_{ij} : \mathrm{int}_{\mathrm{RV}}(A_{ij}) \longrightarrow \mathcal{P}(\mathrm{RV}_\infty)$. Let

$$C = \bigcup_{i,j} C_{ij} \quad \text{and} \quad B = \bigcup_{i,j} \bigcup_{t \in \partial_{\mathrm{RV}} A_{ij}} \mathrm{rv}(A)_t.$$

Obviously $\dim_{\mathrm{RV}}(C) < n$. By the inductive hypothesis, for all A_{ij} we have $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}} A_{ij}) < n-1$. Thus $\dim_{\mathrm{RV}}(B) < n$. Since $\partial_{\mathrm{RV}} A \subseteq B \cup C$, the claim follows. □

Referee R033 [Gap] – RV-boundary proof needs uniform branches.
Original source line 962. When constructing f_{ij}, ensure that each fiber contains exactly one representative of each boundary RV-disc; injectivity of rv on these finite fibers is a hypothesis of Corollary 2.30. The sentence about constancy should mean that the finite set $\mathrm{rv}(f_{ij}(a))$ is independent of $a \in t^\sharp$, not that rv is constant on the union of all those finite sets. The argument also depends on the preceding corollary after its parameter and multidimensional gaps are repaired. A shorter alternative is the human referee's generic-lift argument: over a model containing the parameters, a tuple of full RV-dimension has a lift realizing one complete type, so its lift cannot meet A partially; state the dimension/type fact precisely if using this route.

Let $f : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ be a definable function.

Lemma 2.36 ([17, Lemma 8.7])**.** *There is a definable closed set $A \subseteq \mathrm{VF}^n$ with $\dim_{\mathrm{VF}}(A) < n$ such that $f \upharpoonright (\mathrm{VF}^n \smallsetminus A)$ is continuous.*

Per the usual terminology, this lemma says that f is continuous *almost everywhere*.

Definition 2.37. For any $a \in \mathrm{VF}^n$, we say that f is *differentiable at a* if there is a linear map $\lambda : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ (of VF-vector spaces) such that, for any $\epsilon \in \Gamma$, if $b \in \mathrm{VF}^n$ and $\mathrm{val}(b)$ is sufficiently large then

$$\mathrm{val}(f(a+b) - f(a) - \lambda(b)) - \mathrm{val}(b) > \epsilon.$$

It is straightforward to check that if such a linear function λ (represented by a matrix with entries in VF) exists then it is unique and hence is called the *derivative of f at a*.

Write $f = (f_1, \ldots, f_m)$. For $a = (a_i, a_{\tilde{i}}) \in \mathrm{VF}^n$, if the derivative of the function $f_j \upharpoonright (\mathrm{VF} \times a_{\tilde{i}})$ at a_i exists then we call it the *ijth partial derivative of f at a*.

Referee R034 [Clarification; revised] – Valuation of a tuple and the zero increment.
Original source line 980. Define $\mathrm{val}(b) = \min_i \mathrm{val}(b_i)$ for a nonzero tuple, with $\mathrm{val}(0) = \infty$, and require $b \neq 0$ in the difference quotient. Otherwise the display includes the undefined expression $\infty - \infty$ at $b = 0$. Later, where valuations of tuples are taken coordinatewise (Terminology 2.39), distinguish that use from this minimum convention. The input-first indexing of partial derivatives is legitimate, but please keep the corresponding matrix convention explicit.

We differentiate functions between arbitrary definable sets by "forgetting" the RV-coordinates. More precisely, let

$$f : \mathrm{VF}^n \times \mathrm{RV}^m \longrightarrow \mathrm{VF}^{n'} \times \mathrm{RV}^{m'}$$

be a definable function. By dimension theory, for every $t \in \mathrm{RV}^m$ there is an $s \in \mathrm{RV}^{m'}$ such that $\dim_{\mathrm{VF}}(\mathrm{dom}(f_{(t,s)})) = n$ and hence $\mathrm{dom}(f_{(t,s)})$ has an open subset. For such an $s \in \mathrm{RV}^{m'}$ and each a contained in an open subset of $\mathrm{dom}(f_{(t,s)})$, we define the partial derivatives of f at (a,t) to be those of $f_{(t,s)}$ at a. If $n = n'$ and all the partial derivatives exist at a point (a,t) then the *Jacobian of f at (a,t)* is defined in the usual way (that is, the determinant of the Jacobian matrix) and is denoted by $\mathrm{Jcb}_{\mathrm{VF}} f(a,t)$.

Lemma 2.38. *The Jacobian of f exists and is a continuous function almost everywhere.*